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Auditing Recommendation Bias in LLM-Based Social Media Curation

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— THIS IS ALREADY HAPPENING

LLM curation is live, at scale, today

- On **X**, Grok started ranking over 100M posts a day (both the “For You” and “Following” feeds) by November 2025.
- Around the same time, **Instagram** launched a tool that uses AI to summarize a user’s inferred interests and lets them directly adjust their Reels recommendations.
- In early 2026, **Bluesky** launched **Attie**, an agentic feed builder powered by **Claude**, which is now available for beta users.

— WHY DO WE CARE?

LLM-based rankers may inherit two forms of biases

1. Recommender systems have long exhibited systematic fairness problems, from popularity bias to demographic disparities in exposure

2. LLMs independently carry biases inherited from pre-training corpora and alignment procedures.

— THREE RESEARCH QUESTIONS

What we want to answer

1. **RQ1:** What's the overall bias landscape, and how prompt-sensitive is it?

2. **RQ2:** How do different LLM providers differ in their handling of polarization, sentiment, and content toxicity?

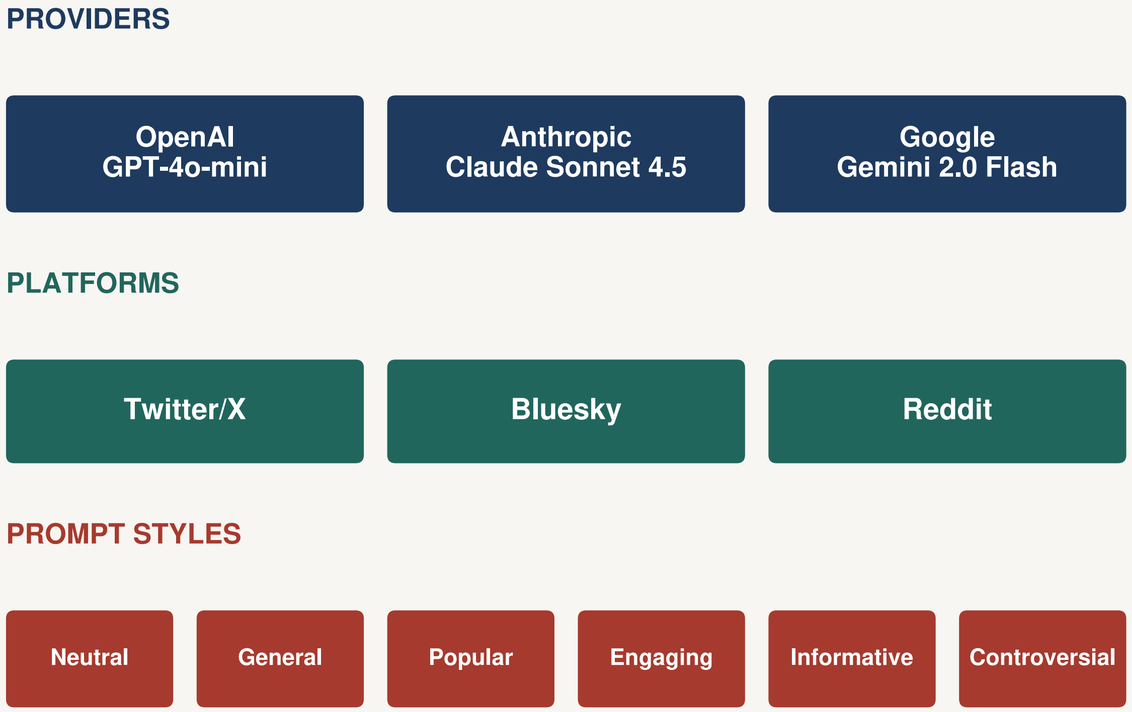
3. **RQ3:** How does bias in gender and political leaning show up, and in which direction?

— AUDIT DESIGN

- **3 Providers**
- **3 Platforms**
- **6 Prompt styles**

54 conditions × 100 trials, recommending top-10 from pools of 100 → 540,000 recommendations.

Note that providers see raw post text only: no metadata, no author signals.



— RQ1 — THE BIAS LANDSCAPE

Polarization dominates every other signal

Polarization is 2× the next-strongest feature.
Then, Primary topic, Toxicity, Political leaning follow.

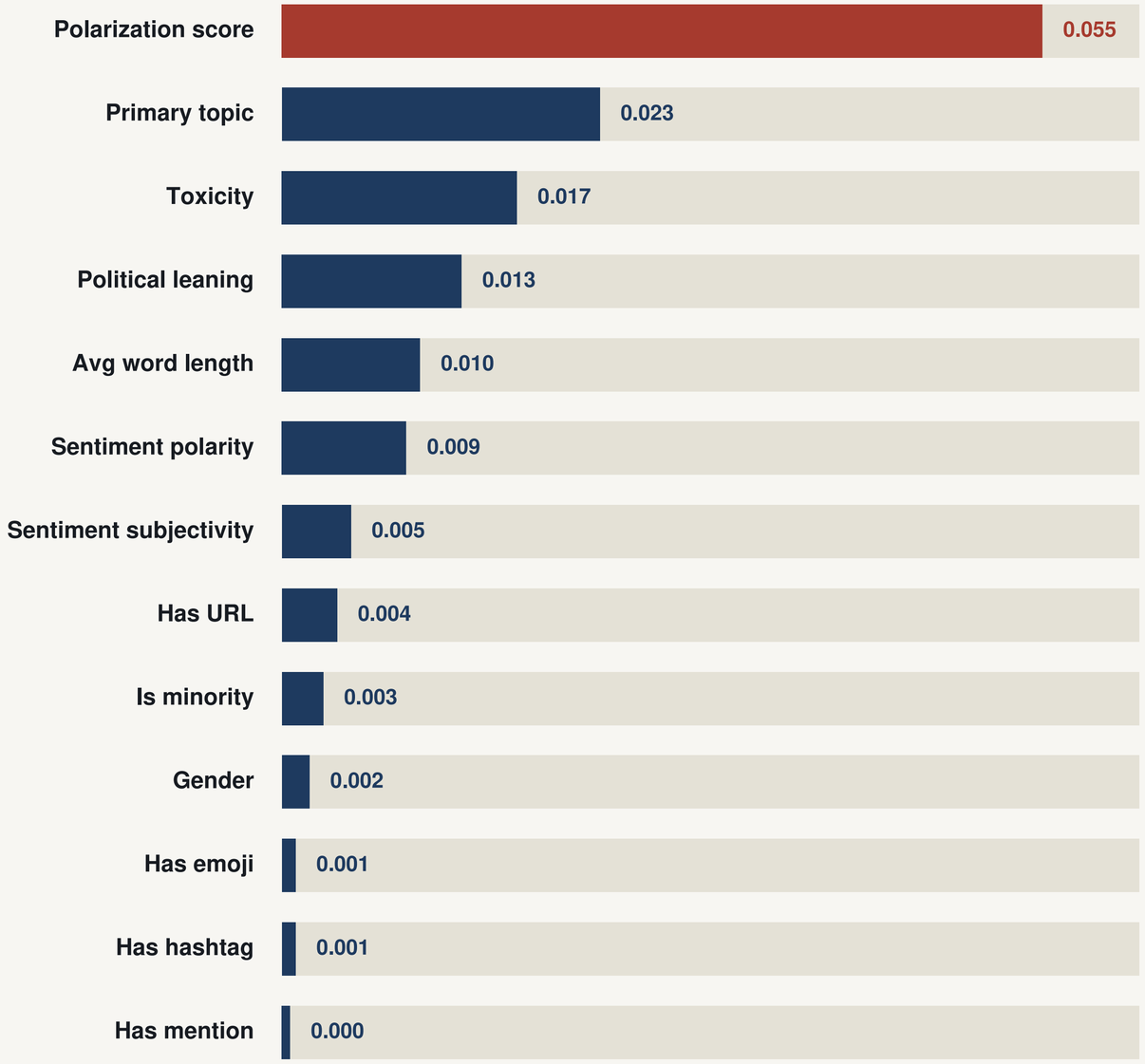


Fig. R^2 (variance explained) for each of the 13 features, averaged across all 54 experimental conditions

Even Neutral is not “neutral”

POLARIZATION

	Claude Sonnet 4.5	GPT-4o mini	Gemini 2.0 Flash	Average
Neutral	+0.188	+0.195	+0.240	+0.208
General	+0.046	+0.086	+0.019	+0.050
Popular	-0.060	+0.066	-0.010	-0.001
Engaging	+0.199	+0.105	+0.166	+0.157
Informative	+0.139	+0.159	+0.148	+0.149
Controversial	+0.252	+0.196	+0.245	+0.231
Average	+0.127	+0.134	+0.135	+0.132

SENTIMENT

	Claude Sonnet 4.5	GPT-4o mini	Gemini 2.0 Flash	Average
Neutral	-0.007	-0.088	-0.126	-0.074
General	+0.022	+0.014	+0.029	+0.022
Popular	-0.009	+0.014	-0.013	-0.003
Engaging	-0.153	-0.027	-0.134	-0.105
Informative	+0.039	+0.003	+0.040	+0.027
Controversial	-0.059	-0.096	-0.130	-0.095
Average	-0.028	-0.030	-0.056	-0.038

TOXICITY

	Claude Sonnet 4.5	GPT-4o mini	Gemini 2.0 Flash	Average
Neutral	-0.058	+0.034	+0.051	+0.009
General	-0.069	-0.021	-0.056	-0.048
Popular	+0.041	+0.010	+0.057	+0.036
Engaging	+0.138	+0.028	+0.120	+0.095
Informative	-0.103	-0.055	-0.101	-0.087
Controversial	-0.032	+0.017	+0.033	+0.006
Average	-0.014	+0.002	+0.017	+0.002

Fig. Content and Safety Directional bias by Model and Prompt style (averaged across platforms)

- Overall, almost all prompts and all providers show **positive polarization bias**, **negative sentiment bias**, and **positive toxicity**.
- Even the **Neutral** prompt exhibits strong polarization, strong negative sentiment, mild toxicity.

— RQ2 — PROVIDER COMPARISON ON PROMPT SENSITIVITY

Providers differ in their level of prompt sensitivity

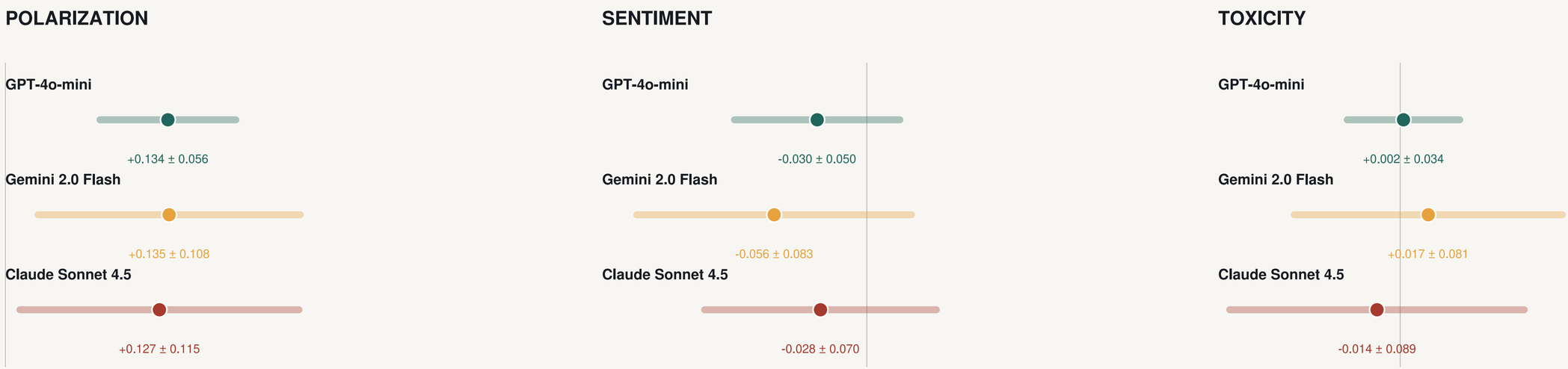


Fig. Mean ± std. of directional bias per provider, aggregated across all 54 conditions

- GPT-4o-mini's spread is consistently the tightest — Claude and Gemini swing much harder depending on the prompt.

— RQ2 — TOXICITY HANDLING

Toxicity: informative vs engaging

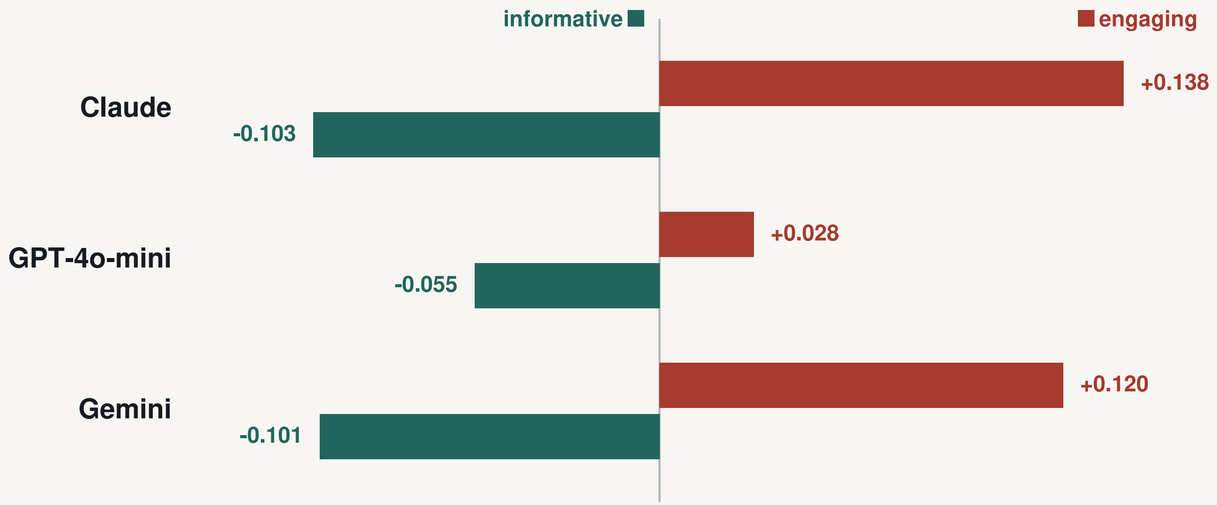


Fig. Directional toxicity bias by provider under the two most divergent prompt styles.

- Under engaging prompts, Claude and Gemini amplify toxicity hard.
- Under informative prompts, all three suppress it. OpenAI stays mild either way.
- Practical takeaway: optimizing for engagement without explicit toxicity constraints has a real cost.

Left-leaning authors crowd the recommendations

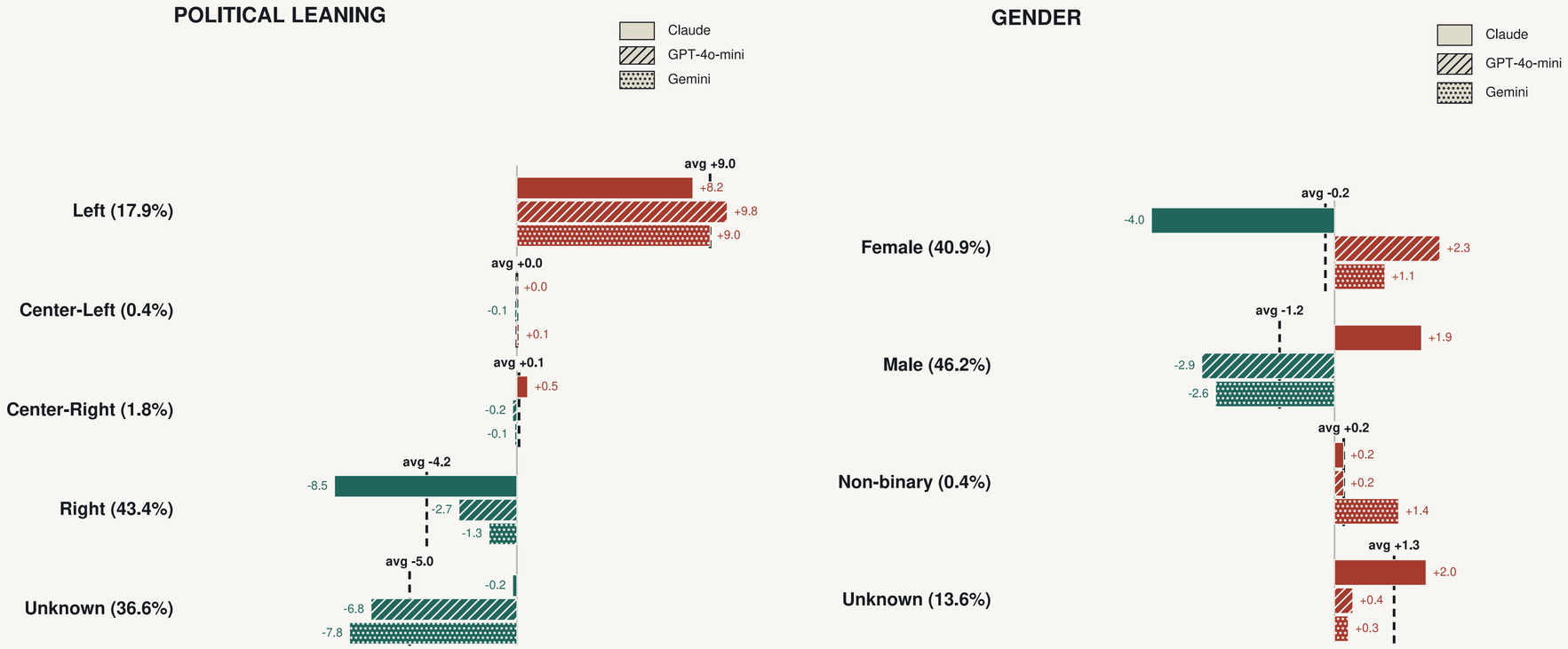


Fig. Directional bias in recommended vs. pool authors, by model, for political leaning and gender.

- Every provider over-represents left leaning authors and under-represents right leaning authors. Moreover, the effect is stronger under controversial, engaging, and even neutral prompt styles.

- Gender is weaker and less consistent. Claude even flips direction relative to the other two.

— WHY DOES THIS HAPPEN?

Indirect demographic skew

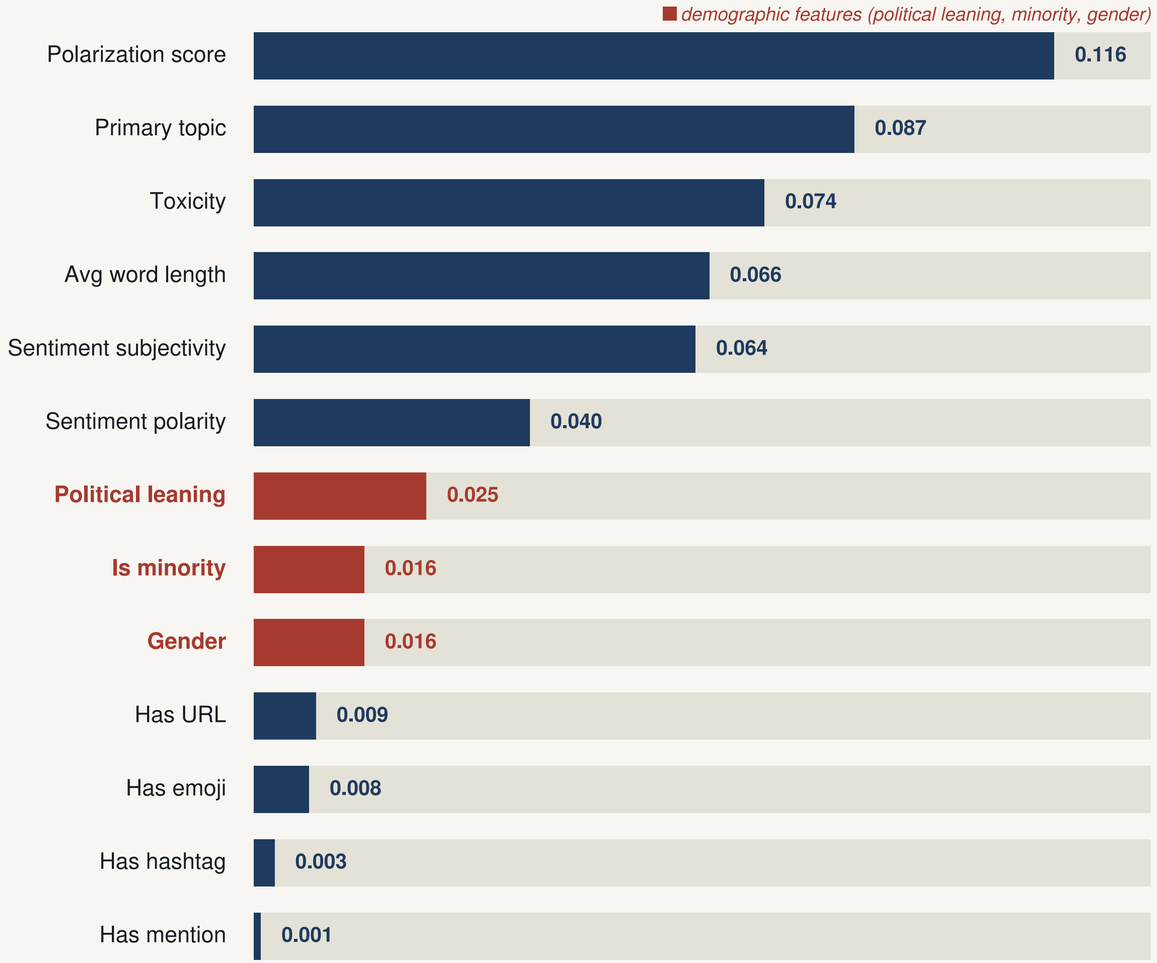


Fig. SHAP feature importance from a Random Forest classifier on recommendation outcomes, averaged across all conditions

A Random Forest on recommendation outcomes (AUROC 0.746) ranks all 13 features. The three demographic features cluster near the bottom, well below every content and even most style features.

The bias depends mostly on correlated content features.

— WHAT THIS MEANS IN PRACTICE

Prompting impacts content bias. Demographic is indirect and mostly prompt-insensitive.

1. **Amplification of extreme content** (more polarized, more negative, and more toxic) is a repeating pattern.

2. **Prompt style** can flip toxicity outcomes, but providers have different prompt-sensitivity.

3. **Left-leaning** authors are consistently **over-represented** (9%), while **Right-leaning** authors are mostly under-represented (-4%). **Demographic skew** is **indirect**: mostly a consequence of content-related preferences and correlation.

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Questions?

nicolo.pagan@uzh.ch · code & data: github.com/paganick/socialmedia_LLM_recsys

Link to Full paper,
published at COLM 2026

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